
Absence Is Not Omission: Availability, Vintage, and the Measurement of Selective Benchmark Disclosure

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Abstract

Model providers choose which benchmark results appear in a release document, while an independent evaluator scores the same models on many more. Whether the left-out results stand below the printed ones seems answerable with a bound on what could have been reported, and benchmark publication dates supply one: an instrument published after a model shipped was not publicly available to it. The comparison this sets up fails as a check on itself. Benchmarks that postdate a release already trail those that predate it by 12.23 percentile points before any disclosure label enters, because the date deciding whether a benchmark could have been reported also fixes where a model sits inside that benchmark’s comparison pool, an age-period-cohort collinearity that reweighting softens but cannot remove. The same failure appears on a leaderboard we did not build: across fourteen HELM Lite releases, a fixed set of models keeps identical published scores while every win rate moves, and a companion leaderboard with a pool-invariant headline shows none of it. Hand-coding every locatable release document, checked by a blind second coding, makes the danger concrete: the coded omission gap is +11.4 points, significant against zero, yet it lands where the label-free gap and the reported share predict. What survives is composition rather than identified concealment: within release, reported benchmarks stand 8.7 points above the omitted eligible ones. No possession-grounded test detects withholding: no document states an evaluation without an outcome, no equal-sized substitution occurs, and the change-based drop gap runs in the predicted direction yet inside its permutation null.

1 Introduction

When a frontier model ships, its provider publishes a model card carrying benchmark results, and independent evaluators run standardised suites over the same models [Liang et al., 2023, Epoch AI, 2026]. This paper asks whether benchmark availability can identify whether the results a provider leaves out stand below the ones it prints, the older economic question. Disclosure unravels only when the sender’s evidence is commonly known [Grossman, 1981, Milgrom, 1981]. Dye [1985] shows what breaks that: a receiver who cannot tell a sender holding nothing from one holding something bad pools the two, and withholding survives. Any study of what was left out inherits that pooling, since the object it needs is the endowment, the results a sender held when it wrote, which the work below does not observe.

Benchmark documentation is one of the few settings where that endowment can be bounded from outside: absent prepublication sharing, a benchmark published after a model shipped

cannot have been in its endowment, whatever a provider ran privately. Dating the instruments therefore separates absence that could have been a disclosure from absence that could not.

Our first contribution is a panel built around that bound. From a pinned snapshot of an independent benchmarking hub covering 819 model-versions, we collapse scaffold variants to the release a provider ships, giving 353 releases across 46 organisations, 74 benchmarks and 3,082 scored pairs. Every benchmark carries a verified release date, splitting the panel into 2,355 eligible pairs and 476 that could not have been disclosed.

Our second contribution is that bounding the endowment is not enough, since judging with-holding needs a measure of how results stood against peers. The eventual comparison is reported versus omitted benchmarks among those available, and it needs the two sides of availability balanced in standing. They are not. Benchmarks that already existed when a model shipped outscore benchmarks built afterwards by 12.23 percentile points on average, positive in 78.1% of the 146 releases holding both kinds of cell, and nothing in that calculation knows which benchmarks a provider reported. The most favourable measure of standing we can construct still leaves 9.14 points. The contamination is structural, because the date identifying the endowment also shifts the measure of standing used to price silence [Heckman, 1979]. Read as economics: instrument vintages are a free verification technology for the endowment that Dye [1985] shows a receiver cannot otherwise see; the verifying variable is not excludable from the receiver’s valuation of silence, so what it verifies it contaminates.

Our third contribution takes the question outside our own panel. Across 14 releases of HELM Lite, the pool grows from 30 models to 91 while 24 models keep bit-identical scores, and every one of their published headline values moves, five pairs reversing order. A companion leaderboard from the same team, behind an absolute headline, shows none of this.

Our fourth contribution codes the record. Hand-coding the 100 releases with a locatable release-time document, blind-replicated on a fifth at $\kappa = 0.95$ on reported versus not, yields the estimator such a study would use. Omitted benchmarks stand +11.4 points above those a provider could not have reported, excluding zero yet short of its matched counterpart by only what composition implies: it cannot separate concealment from the availability gap. What survives is composition: reported benchmarks stand 8.7 points above omitted eligible ones within release ($p = 0.0001$), a gap benchmark fixed effects do not remove.

1.1 Related literature

That reporting in model documentation is uneven, incomplete, and shaped by the sender is established [Staufer et al., 2025, Kuehnert et al., 2026], and two papers approach the omissions themselves. Zhang et al. [2026] examines four flagship releases and finds a benchmark family absent from every main result table; Ghosh et al. [2026] builds the reporting format in which such an absence would become visible. Both work without benchmark dates, so neither can tell a benchmark that was passed over from one the release predates. Bean et al. [2025] reviews 445 benchmarks for construct validity, a complement: it asks whether the instruments measure what they claim; we take them as given and ask which results a provider prints.

On leaderboards the evidence is closer. Singh et al. [2025] documents selection inside a live arena, providers testing private variants and keeping the best, and Xu et al. [2025] moved HELM’s own headline off a win rate that depends on the models compared. Both show rankings moving for reasons other than the evidence; what neither separates, and we make separable at the cell level, is an omission from a result that could not have existed. None of these papers conditions on availability, the single thing we add: without a verified benchmark date, an omission and a nonexistence are the same observation.

The economics supplies the frame and the warning. Silence unravels when the sender’s endowment is commonly known [Grossman, 1981, Milgrom, 1981], and survives when the receiver cannot tell an empty endowment from a concealed one [Dye, 1985, Jung and Kwon, 1988] or when disclosure is costly [Verrecchia, 1983], frictions this design does not separate. The warning holds empirically: unravelling fails in the field [Dranove and Jin, 2010, Jin

and Leslie, 2003, Hotz and Xiao, 2013, Bederson et al., 2018] and in the laboratory [Jin et al., 2021]. What none of this work observes is the endowment itself, the object our bound is built to recover. Concurrent theory models strategic evaluation and persuasion directly [Truong et al., 2026, Dai et al., 2026]; that work is theory, ours measurement.

2 Data and Setting

2.1 The disclosure problem

A release i ships at date t_i carrying a single document. Let $\mathcal{B}$ index benchmarks, each with its own publication date τ_b , let $E_i \subseteq \mathcal{B}$ be the endowment, the benchmarks whose result the provider holds when it writes, and let $D_i \subseteq E_i$ be the disclosed set, which reports a score s_{ib} for each $b \in D_i$. The receiver observes D_i and its scores and must split the complement into results held back and results never held.

Under a commonly known endowment, silence is priced at the worst value it could conceal and disclosure unravels to $D_i = E_i$ [Grossman, 1981, Milgrom, 1981]. Two distinct frictions break that. In Verrecchia [1983] the endowment is known and disclosure is costly, so withholding is a cutoff in s_{ib} . In Section 3 of Dye [1985] disclosure is free but the receiver cannot tell an empty endowment from a concealed one, because a sender can prove possession and cannot prove emptiness.

Here an independent evaluator publishes s_{ib} for cells the provider passed over, and each τ_b makes infeasibility verifiable. Writing $A_i = \{b : \tau_b < t_i\}$ for the available set and O_i for the cells the evaluator scores,

$$D_i \subseteq E_i \subseteq A_i, \quad \theta = \Pr(b \notin D_i \mid b \in O_i \cap A_i).$$

A provider may hold results the evaluator never produced and a benchmark can exist and go unrun, so θ is defined against the evaluator’s coverage rather than the true endowment, and the Dye friction survives inside A_i .

Judging what was withheld then requires a measure of standing P_{ib} , since raw scores are not comparable across benchmarks. The design rests on an exclusion restriction: conditional on the innocent explanations, membership in A_i does not enter the equation for P_{ib} [Heckman, 1979]. A date rule cannot satisfy it: $t_i - \tau_b$, t_i and τ_b are exactly collinear, so if any two of the three move P_{ib} the third cannot be excluded from it. All three move it here, which Section 3 shows and Appendix B bounds. Two things follow, and they are not the same. The first is practical: estimating θ needs disclosure labels, which the coded record of Section 5 supplies. The second is structural: reading θ as evidence of withholding runs into A_i itself. Availability is a deterministic threshold in benchmark maturity, $\mathbf{1}[t_i - \tau_b > 0]$, so no cell of the same maturity sits on both sides of the boundary. Wherever standing depends on maturity, conditioning finely enough to make availability excludable destroys positivity. So availability alone cannot globally identify selective disclosure from any pool-relative standing measure, win rate or percentile alike, and a jointly estimated scale inherits the same selection through which cells exist. What survives, under continuity, is the discontinuity at the threshold.

2.2 The independent score set

The evidence base is a pinned snapshot of Epoch’s benchmarking hub taken on 17 August 2026, covering 819 model-versions. The unit of disclosure is a release, keyed on organisation, model name, and release date. Epoch’s version field splits a release across scaffolds such as reasoning-effort and context-length variants, but a provider ships one document per release, so scaffold rows are collapsed to their release, keeping the best score across variants, spread kept as a diagnostic. The resulting panel holds 353 releases, 74 benchmarks and 3,082 release-benchmark pairs with an independent score. Providers are the unit of dependence.

2.3 Eligibility and benchmark dates

A provider cannot omit a benchmark that did not yet exist. Every benchmark therefore carries its own release date, the earliest verifiable public release, taken from Epoch where

it publishes one and hand-collected from the benchmark’s primary source otherwise, resolving preprint, repository and versioned dates to the earliest public appearance. A pair is eligible only when the benchmark strictly predates the release and its date is verified. The bound is an assumption about access: an instrument shared before publication defeats it. Violations understate rather than manufacture the gap, moving available cells into the postdating group, and the coded record carries no trace of them, no document reporting a benchmark before its public release date. Eligibility yields 2,355 of the 3,082 pairs. Pairs whose benchmark postdates the release form a separate set of 476, the cells that could not have been disclosed. Pairs with no verified date are dropped rather than imputed, and error in the dates themselves attenuates the boundary estimate of Section 3. The release-level contrast imposes no minimum benchmark count, since requiring one would discard exactly the releases whose only cells are a single eligible and a single postdating one. Of the 146 releases carrying both kinds, 57 hold fewer than eight scored benchmarks. A second and simpler obstacle, that the number of benchmarks a release could have reported is itself set by when it shipped, is treated in Appendix A.

Our disclosure-coding protocol uses ORBIT, the Outcome Reporting Bias In Trials instrument from clinical trials, where a protocol-specified outcome vanishing from the published report has a long measurement literature [Chan et al., 2004, Kirkham et al., 2010, Dwan et al., 2013]. Present cells are graded by whether the document gives a number, names the benchmark without one, or substitutes a variant; absent cells by whether a same-family predecessor reported it, contemporaries did, or no comparable provider did either. Of a queue of 108 releases, the 103 with eligible cells in the pinned panel plus five from a later vintage, the 100 with a locatable release-time document are coded under it, 1,280 cells in all. No result before Section 5 rests on a disclosure label.

2.4 Measuring standing

Raw scores are not comparable across benchmarks, so standing is measured as a within-benchmark percentile computed against contemporaneous models, meaning those whose release dates fall within 182 days on either side of the focal release, with midranks for ties. The peer count and the share of the window released after the focal model are retained alongside the percentile. This is the measure a windowed comparison naturally suggests, and the one the paper goes on to show is biased in a way that mimics selective disclosure.

2.5 Inference

Inference is constrained by the same structure. The panel has 46 nominal provider clusters, but 24 contribute fewer than ten cells, 20 appear in a single release, and the five largest hold 68.1% of the sample. Cluster-robust asymptotics are not credible below twelve clusters, and we report no analytic standard error there [Cameron et al., 2008]. Regression coefficients in that range use the restricted wild cluster bootstrap over 9,999 draws, which imposes the null before resampling Rademacher weights at the provider level. Release-level means use a cluster sign-flip randomization test over 9,999 draws, reversing an organisation’s entire contribution at once; it preserves provider dependence but not benchmark dependence, which the two-way clustered regressions carry, and their intervals are percentile bootstraps over 10,000 provider resamples. Both report $(k + 1)/(\text{draws} + 1)$, with k the draws at least as extreme as the observed statistic. Mirrored sign vectors leave 2^{G-1} distinct statistics over G clusters, so a resolution of 0.0001 needs fifteen clusters; the 18 organisations behind the headline estimate clear it. Below twelve clusters the bootstrap switches to a six-point weight distribution whose finer support restores it [MacKinnon and Webb, 2016, 2018].

3 The Availability Gap

3.1 The placebo comparison

Separating results held back from results never held calls for a control group whose endowment is known to be empty [Dye, 1985]. The dates appear to supply one: a benchmark not yet published when the model shipped could not, absent prepublication access, have been

reported, while an eligible one could have been. Whether it was reported is what a disclosure label would say, and none enters here. Whatever makes an eligible benchmark look bad for innocent reasons, difficulty drift, saturation, or a shift in what the field measures, presses on both sets alike, so under those explanations the difference should be zero. One innocent channel is asymmetric by construction: a benchmark published before a model shipped can sit in its training corpus and a later one cannot, so contamination raises eligible standing only, and no reweighting of the comparison can excise it.

It is not zero. Across 146 releases with both kinds of cell, eligible benchmarks outscore postdating ones by 12.23 percentile points on average, with a median of 10.69, and the gap is positive in 78.1% of releases. Nothing in that calculation knows which benchmarks were reported. A measure that balances the two sides of the window, described below, cuts the figure to 9.14 points, positive in 70.9% of 141 releases, without removing it. A cluster sign-flip randomization test over the 18 organisations holding both kinds of cell puts that mean at $p = 0.0001$, the smallest value the draws can return. Cells that could not have been reported already stand lower than cells that could, so the gap is not a control.

3.2 Decomposing the gap

Table 3, in Appendix B, locates the failure, regressing standing on a placebo indicator, equal to one when the benchmark postdates the release, always within release. With release effects alone, the placebo coefficient is -13.529 (2.038). Adding benchmark fixed effects absorbs 63.4% of it and leaves -4.949 (2.183). Benchmark composition therefore carries most of the gap, and structurally so: the panel is unbalanced, and placebo cells concentrate by construction in benchmarks that entered late. Those benchmarks are less saturated: split at median publication date, 8.3% of scores on the later half of the 54 benchmarks with ten or more scores sit in the top decile of the benchmark’s own range against 13.2% on the earlier half ($p = 0.015$). A late instrument therefore leaves more room below the ceiling. Adding the peer-window asymmetry term, the share of a cell’s comparison window released after the focal model, instead of benchmark effects absorbs 39.8% and leaves -8.151 (1.865). Conditioning jointly on benchmark effects and both window terms absorbs all but 0.003 points and leaves -0.003 (1.748), an interval reaching $+3.4$, so the residual is bounded at under a third of the original gap rather than at zero.

The mechanism behind the second dimension is geometric. The comparison window is symmetric in calendar time, 182 days on each side of the release, but a benchmark’s model coverage is not symmetric in calendar time, because coverage begins when the benchmark is built. Figure 3, in Appendix B, plots the two objects this creates against benchmark maturity at release: the share of the window that is newer falls with maturity and jumps at the availability boundary, while standing runs through it without a detectable break. The mechanism is discontinuous where the outcome is not; peer-set composition is a property of timing alone.

The share of a cell’s window released after the focal model is 0.4683 for eligible cells and 0.7065 for placebo cells, against 0.5 for a two-sided window. Within release, the slope of standing on that share is -30.63 percentile points per unit among eligible cells alone. Figure 2, in Appendix B, plots standing against that share for both groups at once. Eligible and placebo cells do not form two clouds at different heights; they trace a single declining curve, common only once benchmark composition is absorbed (Table 3), and differ in where along it they fall. On the variable that assigns them, benchmark maturity at release, they cannot overlap at all, so the contrast is an extrapolation across that boundary rather than a comparison within common support. The check is at the boundary, the one place Section 2.1 leaves identified under continuity [Hahn et al., 2001]. Nothing detectable happens there. Local linear estimates are negative across four bandwidths and two polynomial orders, every interval covers zero though the widest reaches $+8.05$, and a window chosen on covariate balance puts the difference at -2.52 with a randomization p of 0.447 [Cattaneo et al., 2015]. The two groups’ mean maturity differs by 793 days, and the gap lives in that distance rather than at the boundary (grid and balance tests in Appendix B).

Standing decomposes into a weighted average of position among older and newer peers, $\text{percentile} = (1 - w) P_{\text{old}} + w P_{\text{new}}$ with w the newer share of window peers, exact cell by cell.

Replacing the empirical weight with one half defines the side-balanced measure, available for 92.5% of all cells and 92.8% of eligible-or-placebo cells, and cuts the same slope to -8.56 . That the identity holds does not make the gradient an accounting tautology. Shuffling scores within benchmark destroys the capability trend while leaving dates, window membership, peer counts and newer shares bit-identical. Under that null, the slope has mean -0.09 and standard deviation 3.12. The observed -30.63 sits 9.8 standard deviations away, and none of 300 draws reaches it. Replacing scores instead with a within-benchmark linear fit on date plus permuted residuals, so that a capability trend exists and nothing else does, recovers 93% of the observed slope, averaged over 200 draws. Comparison-set geometry is inert on its own and becomes a bias only when it multiplies a real trend in capability.

3.3 Trimming and reweighting

The phenomenon has a name in nonparametric estimation: local averages are biased near the edge of a support, and the standard remedies recentre rather than discard [Gasser and Müller, 1979, Fan, 1992]. What is unusual is the boundary’s coincidence with the endowment: a benchmark was available to a release exactly when the model did not sit at the left edge of that benchmark’s coverage. The indicator of what a provider could have held is therefore the indicator of where the window is one-sided, and removing the edge removes the contrast. The coincidence has an older name, the age-period-cohort identity: release date, benchmark vintage and their difference are exactly collinear, and each has a measured signature here: t_i in the capability trend, τ_b in saturation, $t_i - \tau_b$ in window composition. Only the linear components are unidentified in that structure [Fosse and Winship, 2019a,b], so what fails is a slope. Three sign restrictions bound it: capability rises with the calendar, later benchmarks are less saturated, and standing rises with maturity. All three are assumptions on unidentified components, each motivated by a measurement of an identified quantity. Under them the unidentified component accounts for at most 24 to 34% of the placebo gap across specifications, roughly three percentile points, which is where the saturated regression also lands. The remainder is carried by the identified reduced-form slopes, and Appendix B gives the derivation.

Standing measure	Mean	Median	Positive	Sign-flip p	Releases	Cells
Windowed percentile, $\pm 182d$	+12.23	+10.69	78.1%	0.0001	146	100.0%
Rank against all models ever	+22.75	+22.66	89.0%	0.0001	146	100.0%
Trim to two-sided windows	+9.96	+10.69	72.2%	0.0001	115	70.4%
Side-balanced percentile	+9.14	+9.62	70.9%	0.0001	141	92.8%

Table 1: The label-free placebo null under each candidate measure: release-level mean standing of eligible minus postdating benchmarks, in percentile points. Every entry should be zero. Sign-flip p is the provider-clustered test of Section 2.5, against zero; Cells, the share of cells carrying the measure.

Table 1 shows what follows. Ranking against every model ever scored nearly doubles the contamination; trimming to two-sided windows discards 30% of the cells to remove 19%, and its row is just the uncorrected gap on the retained cells, composition rather than correction. Shortening the window does reduce the gap, to $+8.59$ at 91 days, but in proportion to the asymmetry it removes rather than by breaking the mechanism. Across four widths and four benchmark-count minima the gap is positive in all 16 specifications, from $+8.59$ to $+21.66$, and tracks the eligible-minus-placebo gap in window composition at $r = 0.991$. Dropping any one of the 18 organisations leaves it between $+11.45$ and $+13.20$. The correction is a covariate, not a subsample.

One reading of that number would be too generous to it. Window composition is a function of the same dates that define the placebo indicator, so conditioning on it does not adjust for a confounder sitting outside the comparison; it splits the gap into the channel carrying it. A study that holds disclosure labels and wants an estimate of θ must first argue that window composition is settled before a provider decides what to report, or it will condition away part of what it is measuring. Provider identity explains little of the variance in window composition, which is suggestive rather than conclusive; Appendix B reports the decomposition and the coverage tilt it does not correct.

289 The side-balanced percentile is the one change of measure that does substantial work, cutting
 290 the placebo mean to +9.14 and the eligible-cell gradient to -8.56 . Figure 4 in Appendix B
 291 puts it against the uncorrected measure on the same cells, near flat where cells are dense.
 292 Reweighting cannot reach what remains: balancing repairs the weight on each side of a
 293 model but not which models an evaluator chose to score early, so evaluator attention stays
 294 a residual confound. Undefined wherever a side is empty, it accompanies the uncorrected
 295 measure rather than replacing it.

296 4 Evidence from HELM

297 Everything so far concerns a percentile constructed here on a panel assembled here, so the
 298 effect may belong to the construction rather than to leaderboards in general. HELM Lite
 299 publishes its full accuracy table at every versioned release, archives the earlier ones, and
 300 reports a pool-relative headline [Liang et al., 2023], so the same evidence can be read at 14
 301 points in time while the pool grows underneath it.

302 If a leaderboard moves, the first thing to rule out is a revised suite. Across releases v1.0.0
 303 through v1.13.0, the ten scenario columns are identical, with nothing added, dropped or
 304 renamed beyond one relabelled column header. The same 24 models appear in all 14 releases
 305 with their 240 published scenario scores bit-identical throughout, while the evaluated pool
 306 grows from 30 models to 91.

307 Against that frozen evidence, all 24 constant models see their published Mean win rate
 308 change, by 0.1232 on average in absolute value and by as much as 0.2178 for Mixtral 8x7B.
 309 Most of that movement is arithmetic: Mean win rate is the fraction of head-to-head compar-
 310 isons a model wins, so a fixed model faces a larger, later field as the pool grows; the change
 311 correlates with initial standing at -0.766 .

312 The movement that matters is elsewhere. If the drift were a monotone transform of the
 313 frozen evidence, it would be harmless, since a monotone rescaling preserves every order. It
 314 is not. Of the 24, 22 fall and two rise, and five of the 276 pairs among them, 1.8%, reverse
 315 between the first release and the last, each a published order inversion with bit-identical
 316 evidence beneath it. Figure 1 plots each model’s rank among the constant set at the first
 317 release against the last, so a reversed pair is a crossing, countable on the page while the
 318 evidence beneath is fixed. Unlike test-set reuse [Roelofs et al., 2019], the evidence is fixed;
 319 the reference set is what moves. The published numbers are correct as defined at every
 320 release; what changed was the pool, and the published order followed it. Singh et al. [2025]
 321 reports a comparable pattern from a different mechanism, deprecation policy rather than a
 322 coverage boundary changing which models remain available, and in both the reference set
 323 rather than the evidence moves the ranking.

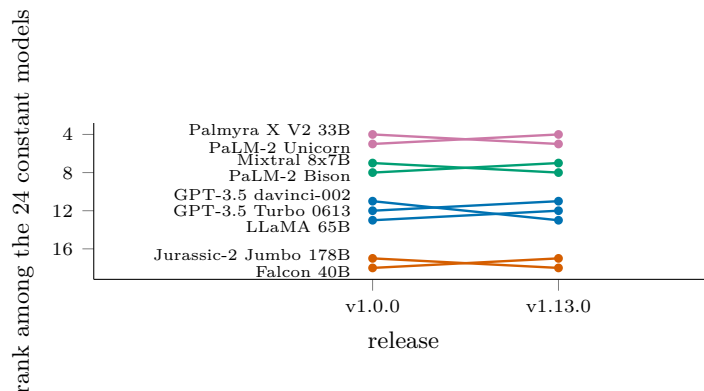


Figure 1: The nine HELM Lite models whose published order changed between the first release and the last, ranked among the 24 whose 240 scenario scores are bit-identical throughout; coloured by group of mutually reversing models, four groups and five reversed pairs in all. The control: across 16 HELM Capabilities releases behind an absolute headline, with the pool growing from 22 to 68, none of the 22 constant models’ headline values moved and no pair reversed.

The control, HELM Capabilities, comes from the same organisation, whose maintainers had diagnosed the win rate as pool-dependent [Xu et al., 2025]; it runs on the same infrastructure and cadence, its pool growing comparably, from 22 models to 68 over 16 releases. Its headline is an absolute mean over scenarios rather than a fraction of comparisons won. The same 22 models appear in every release with 110 scenario scores bit-identical, none of the 22 headline values moves, and no pair reorders, at the endpoints or ever. An absolute mean over bit-identical scores on a fixed suite cannot move with the pool, so zero drift is arithmetic, a scope condition rather than a test that could have failed.

This need not be read as an error on HELM’s part: the absolute headline on its own Capabilities leaderboard is the point. A pool-relative headline is a joint statement about a model and about the population evaluated beside it; read later as statements about the models alone, two such numbers move standing between providers for reasons belonging to neither. Restoring comparability across moments is the classical equating problem [Kolen and Brennan, 2014], avoided by a scale estimated jointly across benchmarks [Ho et al., 2025]. The panel and the leaderboard fail the same way: the reference set, not the evidence, moves the number.

5 Coding the Record

A benchmark can be missing from a model card because it was irrelevant, never run, or run and withheld. Only the third is disclosure in the sense of Grossman [1981], Milgrom [1981] and Dye [1985], and only it is a reader’s to price. The within-family drop narrows it: a provider that reports b at t and omits it at $t + 1$, an independent score present, has established relevance and possession of the harness by its own earlier behaviour, with the independent score as the registry-style counterfactual [Chan et al., 2004, Dwan et al., 2013, Kirkham et al., 2010]. It does not establish that b was run for $t + 1$, so the Dye [1985] uncertainty narrows rather than closes. Its ceiling, 37 pairs, and detection threshold, roughly 38 points at one drop, sit in Appendix B.

A single coder read the release-time document of each of the 100 releases for which one is locatable; eight of the queued 108 have none. The record notes every benchmark a document reports, with variants, bare mentions, and stated benign absences; the ORBIT categories are derived mechanically from these records and the family history. This yields 1,280 coded cells; 31 from five later-vintage releases sit outside the pinned panel and are excluded. A second coder then recoded a provider-stratified fifth of the releases, blind to the first coding and without calibration contact; Appendix B records the one release where that blindness broke. Agreement is Cohen’s κ over the derived cells [Cohen, 1960], with intervals from resampling releases. The two codings agree at 0.95 [0.80, 1.00] on the primitive judgment of whether a benchmark was reported, at 0.85 [0.72, 0.95] over the full category set, and at 0.81 [0.48, 1.00] on the high-versus-low-suspicion split. The single E cell each coding derives is the same cell, encouraging but no reliability estimate for so rare a category, and Appendix B reports the confusion structure. Among the 1,249 eligible coded cells that resolve in the panel, 76.9% report no number, and the release-level mean is 73.1% [65.9, 79.2]. The rate is descriptive and is not, by itself, evidence of selection.

Standing measure	Coded gap	95% CI	No labels	Difference
Windowed percentile, $\pm 182d$	+11.40	[+5.6, +17.0]	+14.13	−2.73
Side-balanced percentile, $\pm 182d$	+7.90	[−0.4, +15.2]	+9.82	−1.92
Rank against all models ever	+22.33	[+18.0, +28.1]	+24.50	−2.17
Windowed percentile, $\pm 90d$	+8.15	[+1.3, +14.7]	+10.36	−2.21
Windowed percentile, $\pm 365d$	+18.03	[+13.9, +23.1]	+20.53	−2.50

Table 2: The coded omission deficit beside the label-free gap, in percentile points, over 71 releases and 13 providers (70 side-balanced). No labels is the same contrast with no disclosure coding, on the same releases; Difference is Coded gap minus No labels. Intervals are provider-clustered bootstraps; four of five coded gaps exclude zero.

Table 2 is the section’s result. The disclosed-versus-omitted comparison returns +8.7 points [4.1, 13.9] over 63 releases; the intended availability contrast, omitted-eligible against post-

367 dating cells over the 71 releases holding both, returns +11.4 [5.6, 17.0] and would reject a
 368 zero null at $p = 0.003$. Read against Section 3, it is the availability gap: the label-free
 369 contrast on the same releases is +14.1, and under every measure and window the coded gap
 370 falls 1.9 to 2.7 points short. The shortfall is an exact accounting identity, the average over
 371 the same releases of the reported share times the within-release reported-omitted gap, +2.7,
 372 so the labels change this contrast only through composition.

373 What survives is composition. Reported benchmarks stand 8.7 points above omitted eligi-
 374 ble ones within release, a gap larger than all 9,999 count-preserving random assignments
 375 ($p = 0.0001$, the draws' floor; null standard deviation 1.9) and one the availability gap can-
 376 not produce. It survives coding-error, variant-reclassification, and leave-one-out exercises
 377 because the availability gap does; benchmark identity does not carry it, since benchmark
 378 fixed effects beside the release effects raise the coefficient from +7.4 to +9.3 and the window-
 379 share control leaves +7.4. It does not identify whether standing drove the reporting: rel-
 380 evance, salience, and within-eligible contamination can move disclosure and performance
 381 together.

382 Possession is the remaining distance; four cuts return one verdict. The drop estimator
 383 realises 16 drops across 15 releases and returns +2.9, opposite the withholding prediction,
 384 sign-flip $p = 0.25$ over five providers, the null its calibration priced. No document states
 385 that a benchmark was evaluated while reporting neither a score nor an outcome, across the
 386 295 cells documenting a run. A change-based test, each dropped benchmark against its
 387 standing under the previous release, gives -3.7, the predicted sign, at permutation $p = 0.16$
 388 over 42 transitions; possession tiers and equal-sized substitutions detect nothing further.
 389 Appendix B details these and three unrepaired limits: unlocatable documents, coverage
 390 that may follow provider emphasis, and the best-across-scaffolds rule.

391 6 Conclusion

392 We set out to measure how much of what providers could report they choose not to. The
 393 coded record returns the number and the reason to distrust it. The contrast is +11.4
 394 points, rejecting zero yet short of its matched label-free counterpart by what composition
 395 implies, and too imprecise to exclude a strategic component of a few points. The record
 396 separates three claims: non-reporting is extensive, composition is performance-related, and
 397 withholding is not established, every possession-grounded test returning null, absent, or
 398 opposite-signed evidence. Dating the instruments bounds the endowment; the bound is
 399 an assumption about access. The same dates fix where a model sits inside a benchmark's
 400 coverage: what identifies the endowment moves the measure pricing silence; reweighting is
 401 a partial defence. Growing the 16-drop record is a reporting-standard question: a jointly
 402 estimated scale [Ho et al., 2025] shrinks the placebo gap without closing it, and carrying
 403 a benchmark set across releases narrows both the empty-endowment excuse and the Dye
 404 [1985] pooling.

405 References

- 406 Andrew M. Bean, Ryan Othniel Kearns, Angelika Romanou, Franziska Sofia Hafner, Harry
 407 Mayne, et al. Measuring what matters: Construct validity in large language model bench-
 408 marks. In *Advances in Neural Information Processing Systems* 38, 2025. arXiv:2511.04703.
- 409 Benjamin B. Bederson, Ginger Zhe Jin, Phillip Leslie, Alexander J. Quinn, and Ben Zou.
 410 Incomplete disclosure: Evidence of signaling and countersignaling. *American Economic*
 411 *Journal: Microeconomics*, 10(1):41–66, 2018. doi: 10.1257/mic.20150178.
- 412 A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements
 413 for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427,
 414 2008. doi: 10.1162/rest.90.3.414.
- 415 Matias D. Cattaneo, Brigham R. Frandsen, and Rocío Titiunik. Randomization inference in
 416 the regression discontinuity design: An application to party advantages in the U.S. senate.
 417 *Journal of Causal Inference*, 3(1):1–24, 2015. doi: 10.1515/jci-2013-0010.

418 An-Wen Chan, Asbjørn Hróbjartsson, Mette T. Haahr, Peter C. Gøtzsche, and Douglas G.
419 Altman. Empirical evidence for selective reporting of outcomes in randomized trials.
420 JAMA, 291(20):2457–2465, 2004. doi: 10.1001/jama.291.20.2457.

421 Jacob Cohen. A coefficient of agreement for nominal scales. Educational and Psychological
422 Measurement, 20(1):37–46, 1960.

423 Yifan Dai, Drew Fudenberg, and Harry Pei. Bayesian persuasion with selective disclosure,
424 2026. arXiv:2601.05914.

425 David Dranove and Ginger Zhe Jin. Quality disclosure and certification: Theory and prac-
426 tice. Journal of Economic Literature, 48(4):935–963, 2010. doi: 10.1257/jel.48.4.935.

427 Kerry Dwan, Carrol Gamble, Paula R. Williamson, and Jamie J. Kirkham. Systematic
428 review of the empirical evidence of study publication bias and outcome reporting bias.
429 PLoS ONE, 8(7):e66844, 2013. doi: 10.1371/journal.pone.0066844.

430 Ronald A. Dye. Disclosure of nonproprietary information. Journal of Accounting Research,
431 23(1):123–145, 1985. doi: 10.2307/2490910.

432 Epoch AI. Ai benchmarking hub. <https://epoch.ai/benchmarks>, 2026. Data snapshot of
433 17 August 2026, CC-BY.

434 Jianqing Fan. Design-adaptive nonparametric regression. Journal of the American Statistical
435 Association, 87(420):998–1004, 1992. doi: 10.1080/01621459.1992.10476255.

436 Ethan Fosse and Christopher Winship. Bounding analyses of age-period-cohort effects. De-
437 mography, 56(5):1975–2004, 2019a. doi: 10.1007/s13524-019-00801-6.

438 Ethan Fosse and Christopher Winship. Analyzing age-period-cohort data: A review
439 and critique. Annual Review of Sociology, 45:467–492, 2019b. doi: 10.1146/
440 annurev-soc-073018-022616.

441 Theo Gasser and Hans-Georg Müller. Kernel estimation of regression functions. In Smoothing
442 Techniques for Curve Estimation, volume 757 of Lecture Notes in Mathematics, pages
443 23–68. Springer, 1979. doi: 10.1007/BFb0098489.

444 Avijit Ghosh, Anka Reuel, Jenny Chim, Wm. Matthew Kennedy, et al. Evaluation cards:
445 An interpretive layer for ai evaluation reporting, 2026. arXiv:2606.09809.

446 Sanford J. Grossman. The informational role of warranties and private disclosure about
447 product quality. Journal of Law and Economics, 24(3):461–483, 1981. doi: 10.1086/
448 466995.

449 Jinyong Hahn, Petra Todd, and Wilbert Van der Klaauw. Identification and estimation
450 of treatment effects with a regression-discontinuity design. Econometrica, 69(1):201–209,
451 2001. doi: 10.1111/1468-0262.00183.

452 James J. Heckman. Sample selection bias as a specification error. Econometrica, 47(1):
453 153–161, 1979. doi: 10.2307/1912352.

454 Anson Ho, Jean-Stanislas Denain, David Atanasov, Samuel Albanie, and Rohin Shah. A
455 rosetta stone for AI benchmarks, 2025. arXiv:2512.00193.

456 V. Joseph Hotz and Mo Xiao. Strategic information disclosure: The case of multiattribute
457 products with heterogeneous consumers. Economic Inquiry, 51(1):865–881, 2013. doi:
458 10.1111/j.1465-7295.2010.00340.x.

459 Ginger Zhe Jin and Phillip Leslie. The effect of information on product quality: Evidence
460 from restaurant hygiene grade cards. Quarterly Journal of Economics, 118(2):409–451,
461 2003. doi: 10.1162/003355303321675428.

462 Ginger Zhe Jin, Michael Luca, and Daniel Martin. Is no news (perceived as) bad news?
463 an experimental investigation of information disclosure. American Economic Journal:
464 Microeconomics, 13(2):141–173, 2021. doi: 10.1257/mic.20180217.

465 Woon-Oh Jung and Young K. Kwon. Disclosure when the market is unsure of information
466 endowment of managers. *Journal of Accounting Research*, 26(1):146–153, 1988.

467 Jamie J. Kirkham, Kerry M. Dwan, Douglas G. Altman, Carrol Gamble, Susanna Dodd,
468 Rebecca Smyth, and Paula R. Williamson. The impact of outcome reporting bias in
469 randomised controlled trials on a cohort of systematic reviews. *BMJ*, 340:c365, 2010. doi:
470 10.1136/bmj.c365.

471 Michael J. Kolen and Robert L. Brennan. *Test Equating, Scaling, and Linking: Methods
472 and Practices*. Springer, 3rd edition, 2014. doi: 10.1007/978-1-4939-0317-7.

473 Blaine Kuehnert, Nari Johnson, Ravit Dotan, and Hoda Heidari. Disclosure or mar-
474 keting? analyzing the efficacy of vendor self-reports for vetting public-sector ai. In
475 *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2026. doi:
476 10.1145/3805689.3812220.

477 Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic evaluation of language models.
478 *Transactions on Machine Learning Research*, 2023. arXiv:2211.09110.

479 Wei Ma and George H. Chen. Missing not at random in matrix completion: The effectiveness
480 of estimating missingness probabilities under a low nuclear norm assumption. In *Advances
481 in Neural Information Processing Systems 32*, 2019. arXiv:1910.12774.

482 James G. MacKinnon and Matthew D. Webb. Wild bootstrap inference for wildly different
483 cluster sizes. *Journal of Applied Econometrics*, 32(2):233–254, 2016. doi: 10.1002/jae.
484 2508.

485 James G. MacKinnon and Matthew D. Webb. The wild bootstrap for few (treated) clusters.
486 *The Econometrics Journal*, 21(2):114–135, 2018. doi: 10.1111/ectj.12107.

487 Justin McCrary. Manipulation of the running variable in the regression discontinuity design:
488 A density test. *Journal of Econometrics*, 142(2):698–714, 2008. doi: 10.1016/j.jeconom.
489 2007.05.005.

490 Paul R. Milgrom. Good news and bad news: Representation theorems and applications.
491 *Bell Journal of Economics*, 12(2):380–391, 1981. doi: 10.2307/3003562.

492 Rebecca Roelofs, Vaishaal Shankar, Benjamin Recht, Sara Fridovich-Keil, Moritz Hardt,
493 John Miller, and Ludwig Schmidt. A meta-analysis of overfitting in machine learning. In
494 *Advances in Neural Information Processing Systems 32*, 2019.

495 Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D’Souza, Sayash Kapoor, Ahmet Üstün,
496 Sanmi Koyejo, Yuntian Deng, Shayne Longpre, Noah A. Smith, Beyza Ermiş, Marzieh
497 Fadaee, and Sara Hooker. The leaderboard illusion. In *Advances in Neural Information
498 Processing Systems 38, Datasets and Benchmarks Track*, 2025. arXiv:2504.20879.

499 Sandip Sinharay and Paul W. Holland. The missing data assumptions of the NEAT design
500 and their implications for test equating. *Psychometrika*, 75(2):309–327, 2010. doi: 10.
501 1007/s11336-010-9156-6.

502 Leon Stauffer, Mick Yang, Anka Reuel, and Stephen Casper. Audit cards: Contextualizing
503 ai evaluations, 2025. arXiv:2504.13839.

504 Sang Truong, Serena Wang, Nick Haber, and Sanmi Koyejo. Strategic evaluation: Incen-
505 tivizing AI capability coverage with private benchmarks, 2026. Working paper.

506 Robert E. Verrecchia. Discretionary disclosure. *Journal of Accounting and Economics*, 5:
507 179–194, 1983. doi: 10.1016/0165-4101(83)90011-3.

508 Jialiang Xu, Yifan Mai, and Percy Liang. HELM capabilities: Evaluating LMs capability
509 by capability. *Stanford Center for Research on Foundation Models blog*, 2025. URL
510 <https://crfm.stanford.edu/2025/03/20/helm-capabilities.html>.

511 Chuyifei Zhang, Hongyu Cui, Xiaowen Huang, and Jitao Sang. Positional failures in long-
512 context LLMs: A blind spot in reasoning benchmarks, 2026. arXiv:2605.23170.

513 A What a Provider Could Report

514 The set of results a provider could publish is not fixed, and it grows. When we count the
 515 benchmarks in our panel by the year each was published, the dated stock rises from 11 at
 516 the end of 2019 to 62 by 2026. A benchmark that enters that stock stays available to every
 517 release after it, so the denominator of any omission rate is weakly increasing by construction.

518 What a single release could report grows more slowly and less evenly. Averaged by half-year,
 519 the number of eligible benchmarks per release runs 6.8 in the first half of 2023 and falls to
 520 3.9 in the first half of 2024, a period in which releases outpaced the instruments now used
 521 to score them. By the first half of 2026 it reaches 10.1. The two series move together over
 522 the sample and apart within it, which is consistent with the opportunity set being shaped
 523 partly by release timing rather than by provider choice.

524 A count of omissions therefore measures the calendar. Unless reported tables expand one
 525 for one with the stock, omission rates drift upward for arithmetic reasons alone, and com-
 526 paring early releases with late ones on that measure compares eras rather than providers.
 527 The question that survives is composition: which benchmarks are dropped, conditional on
 528 availability, against what an indifferent selection over the same set would have produced.
 529 Answering it requires knowing not only what was available but how the release stood on
 530 each available benchmark, and it is that second requirement, not the first, that the rest of
 531 the paper shows to be the hard one [Dranove and Jin, 2010].

532 B Additional Checks

533 Is window composition a provider’s choice? Section 3.3 conditions on window composition
 534 and notes that doing so is defensible only if composition is settled before a provider decides
 535 what to report. Regressing the share of a cell’s window released after the focal model on
 536 provider dummies gives an R^2 of 0.071 across 45 organisations, against 0.002 for benchmark
 537 dummies over 61 benchmarks and 0.104 for the release date alone. Window composition is
 538 close to idiosyncratic at the cell level, which is consistent with its being set by the evaluator’s
 539 scoring schedule rather than by the provider, and is not enough to establish it.

540 Is the evaluator’s coverage selective? The estimand conditions on the cells the evaluator
 541 scores, so coverage is upstream of everything else. Coverage averages 8.7 benchmarks per
 542 release with a standard deviation of 8.0 and a range of 1 to 38. It correlates with a provider’s
 543 release count at 0.203 and with the release date at 0.265, and releases from the densest
 544 quartile of providers carry 9.6 scored benchmarks against 8.4 for the rest. Coverage is
 545 therefore mildly tilted toward larger and later providers, and nothing in this paper corrects
 546 for that.

547 Bounding the unidentified component. Write t for a release date, τ for a benchmark’s
 548 vintage and $a = t - \tau$ for the benchmark’s maturity when the model met it. A regression
 549 of standing on all three is singular by construction, which is the age-period-cohort problem,
 550 and the data identify only $\Pi_p = \beta_a + \beta_t$ and $\Pi_c = \beta_\tau - \beta_a$, leaving a one-parameter family
 551 indexed by $\lambda = \beta_a$. Fitting the reduced form on the 2,831 comparable cells gives $\Pi_p = +4.12$
 552 and $\Pi_c = -1.36$ percentile points per year.

553 Three restrictions come from the paper rather than from outside it. Capability rises over
 554 calendar time, so $\beta_t \geq 0$. Later benchmarks are less saturated, which Section 3.2 measures
 555 directly, so $\beta_\tau \leq 0$. Standing rises with maturity, since a model on a young benchmark is
 556 ranked against newer peers, so $\beta_a \geq 0$. Together these give $0 \leq \lambda \leq \min(\Pi_p, -\Pi_c) = 1.36$,
 557 and saturation rather than capability is the binding restriction. The mean maturity gap
 558 between eligible and placebo cells is 2.17 years, so the unidentified component can account
 559 for at most $1.36 \times 2.17 = 2.94$ percentile points, 24% of the observed gap. Adding centred
 560 quadratic and cubic terms in t and τ , which are not collinear with a , moves the ceiling to
 561 34% and 31%.

562 Shuffling scores within benchmark leaves every date and the rank deficiency untouched while
 563 destroying the capability trend. Under that null the same quantity falls to 0.02 percentile

564 points on average over 30 draws, with a maximum of 0.24, so the bound reflects the trend
 565 the geometry multiplies rather than the geometry alone.

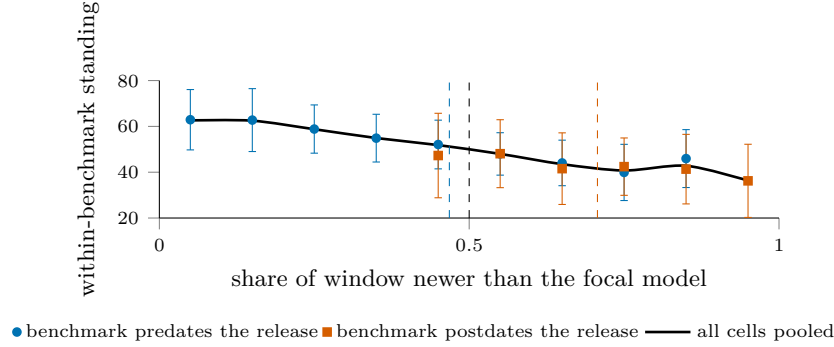


Figure 2: Standing against the share of a cell’s comparison window released after the focal model; dashed verticals mark each group’s mean share, and the black dashed line the symmetric 0.5. Eligible and placebo cells lie on one declining curve, so the contrast between them is a position on that curve rather than a fact about disclosure. Intervals are clustered on organisation.

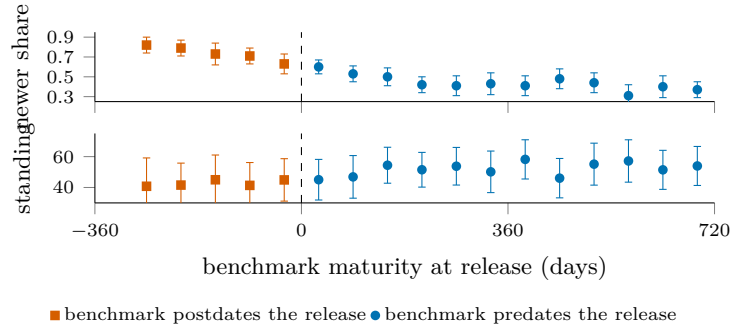


Figure 3: The mechanism and the outcome at the availability boundary. Top: the share of a cell’s comparison window released after the focal model, against benchmark maturity at release; it falls with maturity and jumps at the boundary. Bottom: within-benchmark standing on the same axis, continuous through it. Sixty-day bins; intervals clustered on organisation.

566 The boundary as a discontinuity. Eligibility is $\mathbf{1}[t_i > \tau_b]$, a deterministic function of the
 567 two dates, and that is what the global repairs run into. Equating across groups that took
 568 different instruments assumes a conditional relationship common to both, which two groups
 569 defined by the boundary cannot have [Sinharay and Holland, 2010], and reweighting by
 570 the probability a cell is observed needs that probability bounded away from zero, which a
 571 deterministic rule denies [Ma and Chen, 2019]. The same determinism makes the boundary a
 572 sharp discontinuity, where standing at the threshold is identified under continuity whatever
 573 happens globally [Hahn et al., 2001]. Local linear estimates with a triangular kernel and
 574 provider-clustered errors give -8.82 (4.76) at ± 60 days, -6.85 (4.09) at ± 90 , -4.08 (3.11)
 575 at ± 180 and -2.85 (2.35) at ± 250 , and a quadratic fit at each width moves none of them
 576 across zero. All eight are negative, all eight intervals cover zero, and the largest upper bound
 577 is $+8.05$, below the $+12.23$ the global comparison reports.

578 The local randomization framework agrees [Cattaneo et al., 2015]. Widening symmetric
 579 windows and testing the release’s benchmark count, scaffold count and scaffold spread for
 580 balance, we find the smallest of the three p -values to be 0.335 at ± 20 days and 0.571
 581 at ± 40 before falling to 0.058 at ± 60 , which selects ± 40 , a rule applied after seeing the
 582 balance results, so the randomization p below does not adjust for the selection. The same
 583 difference is -5.96 ($p = 0.23$) at ± 20 days and $+0.43$ ($p = 0.88$) at ± 60 . That window holds
 584 113 postdating and 114 eligible cells, and the difference in mean standing is -2.52 with a
 585 randomization p of 0.447 . Cell counts in the 30 days either side of the cutoff stand at 80
 586 against 84 , a ratio of 0.95 , so the running variable is not bunching against the threshold
 587 [McCrary, 2008].

588 The mechanism is discontinuous where standing is not. The share of a cell’s window released
589 after the focal model jumps +0.083 at the cutoff, which at the eligible-cell gradient of -30.63
590 implies about -2.5 percentile points, inside every interval above, so at the cutoff the design
591 separates concealment from the availability gap only above that effect, and the narrowest
592 windows cannot reject the full -12.23 either. Peer count and raw score are continuous.
593 The global gap therefore lives in the 793 days that separate the two groups’ mean maturity
594 rather than in anything happening at the boundary itself.

Conditioning set	Placebo coefficient	t	Absorbed
Release fixed effects only	-13.529 (2.038)	-6.637	0.0%
Release FE + benchmark fixed effects	-4.949 (2.183)	-2.267	63.4%
Release FE + peer-window asymmetry	-8.151 (1.865)	-4.370	39.8%
Release FE + peer count	-2.107 (1.882)	-1.120	84.4%
Two-way fixed effects + both window terms	-0.003 (1.748)	-0.002	100.0%

Table 3: Placebo coefficient by conditioning set, in percentile points, always within release. Two-way provider-by-benchmark standard errors in parentheses (45 by 61 clusters; provider-only in Appendix B). $n = 2,831$ cells. Absorbed is the fall in the absolute coefficient from the first row; peer count alone is a suppressor, so the last row conditions on both window terms jointly.

595 A standing measure with no comparison window. The conclusion points at a scale esti-
596 mated jointly across benchmarks rather than against contemporaries, which removes the
597 comparison window from the measure entirely. Fitting one does not remove the gap. Abil-
598 ity and difficulty are estimated together over the 63 benchmarks reported on a common
599 zero-to-one scale, on a logit link, with the loading ridged toward one because a free load-
600 ing is unidentified at this density and takes slopes past 65,000. Restricting to benchmarks
601 carrying at least twenty releases and releases carrying at least three benchmarks leaves
602 1,669 eligible cells over 31 benchmarks. The model is fit on eligible cells alone, so the 226
603 postdating cells it can score are out of sample.

604 The eligible-minus-placebo gap in the residuals is +0.336 residual standard deviations,
605 against +0.449 for the windowed percentile on the same contrast, positive in 64.0% of
606 the 89 releases holding both kinds, at $p = 0.005$. Removing the window shrinks the gap by
607 about a quarter and leaves it positive, which puts a jointly estimated scale beside trimming
608 at 19% and side-balancing at 25%, each on its own scale, rather than ahead of them. Its
609 difficulty parameters are estimated from the maturity-selected cells that exist, so part of the
610 surviving gap may be the same selection re-expressed, and its placebo cells are scored out of
611 sample while every eligible cell is in sample, an asymmetry the contrast inherits. Shuffling
612 scores within benchmark, which leaves every date and the rank deficiency untouched, drops
613 the same quantity to 0.02 on average over 30 draws.

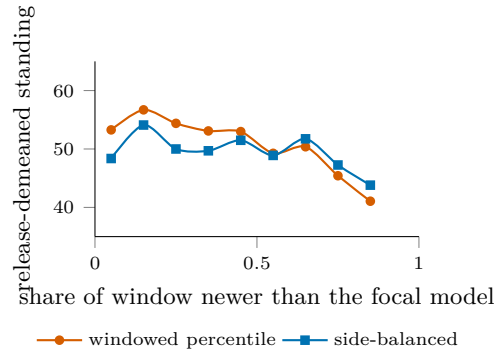


Figure 4: The side-balanced percentile against the windowed one, on eligible cells within release. The windowed measure slopes across the range while the side-balanced one is close to flat where cells are dense; the quoted gradients, -30.63 to -8.56 per unit share, are the within-release regression slopes on these cells, and the release-demeaned binned means shown are correspondingly flatter.

614 A backward-only standing, position among older peers alone, removes the window's forward
615 half entirely and leaves the label-free contrast at +12.11 across 142 releases, so the gap does
616 not come from look-ahead.

617 Drop-design ceiling and calibration. The panel contains 144 adjacent same-family release
618 pairs covering 711 shared eligible benchmarks and 15 organisations. Requiring both releases
619 in a pair to carry at least eight scored benchmarks leaves 37 pairs, 482 shared benchmarks,
620 and 8 organisations, the effective cluster count, in the regime where Section 2.5 reports no
621 analytic standard error. Under the null of no selective withholding, the sampling standard
622 deviation of a drop gap, over 400 draws for each of 117 qualifying releases, is 19.4 percentile
623 points when one benchmark is dropped, 14.4 when two are and 12.4 when three are. At
624 conventional two-sided size a single-drop event must therefore move the gap by roughly 38
625 points before it separates from noise, three times the contamination itself, and no single
626 drop can be read alone.

627 Possession evidence. Selective withholding is a statement about results a provider held and
628 did not print, and the disclosure label observes only the printing. Four cuts organise how far
629 the record reaches toward possession. The direct cases first: no document explicitly states
630 that a benchmark was evaluated while reporting neither a score nor an outcome, so the
631 direct-withholding category is empty over the 295 cells where a run is documented; the six
632 named-without-number cells all carry an outcome in a figure, a chart, or a comparative claim.
633 Second, the omission deficit stratified by evidence tier. Withholding predicts the deficit
634 weakens, and eventually inverts, as possession evidence strengthens; composition predicts
635 no gradient. The point estimates are +18.9 over 21 releases for dropped benchmarks, +11.3
636 over 51 for peer-reported omissions, and +11.3 over 69 for silent ones, The deficit does not
637 become more negative as possession evidence strengthens; the dropped tier instead carries
638 the largest positive deficit, a gradient that does not match the signature withholding predicts.
639 Third, a change-based drop test. The estimator above compares levels; a sharper question
640 is whether a benchmark is dropped after the family's standing on it deteriorates, which
641 differences out persistent salience and relevance. Over the 42 family transitions with a scored
642 predecessor report, 33 dropped and 86 retained cells give a dropped-minus-retained change
643 of -3.7 percentile points, the sign withholding predicts, at a count-preserving permutation
644 $p = 0.16$ within transitions; on raw scores, kept where both sit on the unit interval, the
645 gap is -0.7 points at $p = 0.71$. Both samples cut the 63 derived drop cells: the level
646 estimator keeps the 16 sitting in releases that also retain at least two reported benchmarks
647 to compare against, while the change-based test keeps the 33 with an independent score
648 under both predecessor and successor and imposes no retention floor. Fourth, substitutions.
649 Requiring the reporting table to hold its size while one benchmark is dropped and another
650 inserted, the case a fixed-table explanation cannot cover, leaves no qualifying case in the
651 record. Nothing in the four detects withholding. The change-based gap is directionally
652 consistent with it and statistically indistinguishable from its permutation null; every other
653 cut is null, absent, or opposite in sign.

654 Reliability detail. The second coding covers 20 of the 100 readable releases, drawn strat-
655 ified by provider. Derived categories import contemporaries' evidence across releases, a
656 dependence the release resampling does not model. Across 15 of them, 194 derived cells
657 pair: one drawn release is a zero-disclosure document whose blank evidence rows the pair-
658 ing drops, a conservative exclusion since all eleven of its cells are trivial agreements, and
659 four drawn releases sit outside the pinned panel. The reported-versus-not cross-tabulation
660 is one-directional, 47 jointly reported and 143 jointly not: no cell the first coder called re-
661 ported was coded otherwise by the second, and the four reversals all sit on one release whose
662 second-coding note records retaining four contested scores despite the first coder's disagree-
663 ment, an exposure that breaks blindness for that release. Excluding it, reported-versus-not
664 agreement is exact and the full-category κ is 0.88 on 182 cells. Of the 14 disagreements
665 overall, seven are the variant boundary between C and A, one a bare mention against a
666 value, two the derived split between G and H on a single release, and four the exposed
667 release above. Three limits of the record are candidly unrepaired: the eight releases without
668 a locatable document are unlikely to be missing at random and plausibly document less; the
669 evaluator's coverage may itself respond to what providers emphasise, which would tilt the

670 reported set upward by sampling; and the best-across-scaffolds score rule is not probed for
671 asymmetry across the boundary. High-suspicion marginals match exactly at 5.7% for each
672 coder, which is why the split κ holds at 0.81 despite the category's rarity; its wide interval
673 reflects eleven such cells concentrating in a few releases.

674 Provider-only clustering, for comparison. The manuscript's regressions cluster on provider
675 and benchmark jointly. Provider-only errors are consistently smaller: 1.271 against 2.038
676 on the release-effects row, a ratio of 1.60, and 1.652 against 1.748 on the fully saturated
677 row, a ratio of 1.06. The benchmark dimension carries real dependence on the raw contrast
678 and almost none once benchmark effects and the window terms are in, which is what the
679 decomposition says it should. The headline sign-flip test at $p = 0.0001$ preserves provider
680 dependence only; the two-way regression row above is the statement that the contamination
681 survives both dimensions, at $t = -6.6$.

682 NeurIPS Paper Checklist

683 1. Claims

684 Question: Do the main claims made in the abstract and introduction accurately
685 reflect the paper’s contributions and scope?

686 Answer: [\[Yes\]](#)

687 Justification: The abstract and introduction state both findings within their identi-
688 fied scope: the coded omission gap is indistinguishable from the label-free gap, and
689 within-release composition tracks standing. Section 5 reports the reliability of the
690 coding behind them.

691 2. Limitations

692 Question: Does the paper discuss the limitations of the work performed by the
693 authors?

694 Answer: [\[Yes\]](#)

695 Justification: Limitations appear where they bear on the argument rather than
696 only at the end. Section 2.1 bounds the estimand and states that the reported
697 omission rate is descriptive and identifies no withholding; Section 3.2 states that
698 the conditioning is a decomposition rather than a control strategy, and what a
699 study holding disclosure labels would additionally have to argue; Section 3.3 reports
700 the corrections that fail and the coverage the balanced measure loses; Section 5
701 reports the ceiling on the design that narrows endowment uncertainty furthest, its
702 16 realised drops, and the underpowered null they return.

703 3. Theory assumptions and proofs

704 Question: For each theoretical result, does the paper provide the full set of assump-
705 tions and a complete (and correct) proof?

706 Answer: [\[Yes\]](#)

707 Justification: Section 2.1 states one identification result, that a temporal eligibility
708 gate cannot satisfy the exclusion restriction the design needs, and gives its argument
709 in full: the three date variables are exactly collinear, so excluding one requires
710 that at least two of them not move standing. Section 3 shows all three move
711 it. Appendix B states the three sign restrictions used to bound the unidentified
712 component and derives the bound from them. Everything else in Section 2.1 restates
713 published results with citation.

714 4. Experimental result reproducibility

715 Question: Does the paper fully disclose all the information needed to reproduce
716 the main experimental results of the paper to the extent that it affects the main
717 claims and/or conclusions of the paper (regardless of whether the code and data
718 are provided or not)?

719 Answer: [\[Yes\]](#)

720 Justification: Every number in the paper is written by one command into a single
721 machine-readable file, and the data snapshot it was computed against is pinned
722 by a per-file SHA-256 manifest in the released code. Section 2 states the release
723 key, the eligibility rule, the window width, and which estimates impose a minimum
724 benchmark count and which do not.

725 5. Open access to data and code

726 Question: Does the paper provide open access to the data and code, with sufficient
727 instructions to faithfully reproduce the main experimental results, as described in
728 supplemental material?

729 Answer: [\[Yes\]](#)

730 Justification: Code and derived data are released with instructions. The indepen-
731 dent scores are redistributed by their publisher under CC-BY and are fetched rather
732 than copied, then verified against the pinned manifest, so an upstream change is
733 reported rather than silently absorbed. The external leaderboard evidence is frozen

with a SHA-256 per source file, because a public leaderboard moves and the argument depends on the exact values.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: There is no model training. The choices that affect the estimates are stated in Section 2: the release key, the rule for collapsing scaffold variants, the 182-day comparison window, which estimates impose a minimum benchmark count and which do not, and the treatment of pairs whose benchmark vintage is unverified.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Section 2.5 states the inference protocol. The cluster count is small and unevenly distributed, so below twelve clusters no analytic standard error is reported, coefficients in that range use a restricted wild cluster bootstrap, release-level means use a cluster sign-flip randomization test, and no result is called significant on a cluster-t alone. Table 1 reports a standard error with every coefficient, the headline contrast carries a randomization p-value, and the permutation null is reported with its mean, its spread and the number of draws reaching the observed value.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All analysis runs in seconds on a laptop. There is no training, no accelerator use, and no discarded run.

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses published benchmark scores and public release documents, involves no human subjects and no personal data, and makes no claim about any individual.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 4 states the consequence for a reader of a pool-relative leaderboard, namely that such a number describes standing within one vintage of the evaluated pool rather than a property of the model. The conclusion states that whether the record needed to measure withholding should grow is a reporting-standard question. The paper does not attribute concealment to any named provider, because the design does not support that.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

785 Answer: [N/A]
786 Justification: No model, dataset or other asset with misuse potential is released.
787 The release is analysis code and derived tables built from already public sources.

788 12. Licenses for existing assets
789 Question: Are the creators or original owners of assets (e.g., code, data, models),
790 used in the paper, properly credited and are the license and terms of use explicitly
791 mentioned and properly respected?
792 Answer: [Yes]
793 Justification: The independent scores are credited to their publisher and used under
794 CC-BY, and the external leaderboard is cited to its published description. Both are
795 named in the text and in the released code.

796 13. New assets
797 Question: Are new assets introduced in the paper well documented and is the
798 documentation provided alongside the assets?
799 Answer: [Yes]
800 Justification: The released code documents how each number is produced, the snap-
801 shot manifest records the exact data build, and the frozen external evidence records
802 a source URL, a retrieval date and a SHA-256 for every file.

803 14. Crowdsourcing and research with human subjects
804 Question: For crowdsourcing experiments and research with human subjects, does
805 the paper include the full text of instructions given to participants and screenshots,
806 if applicable, as well as details about compensation (if any)?
807 Answer: [N/A]
808 Justification: The work involves no crowdsourcing and no human subjects.

809 15. Institutional review board (IRB) approvals or equivalent for research with human
810 subjects
811 Question: Does the paper describe potential risks incurred by study participants,
812 whether such risks were disclosed to the subjects, and whether Institutional Review
813 Board (IRB) approvals (or an equivalent approval/review based on the requirements
814 of your country or institution) were obtained?
815 Answer: [N/A]
816 Justification: The work involves no human subjects, so no review board approval
817 was required.

818 16. Declaration of LLM usage
819 Question: Does the paper describe the usage of LLMs if it is an important, original,
820 or non-standard component of the core methods in this research? Note that if the
821 LLM is used only for writing, editing, or formatting purposes and does not impact
822 the core methodology, scientific rigor, or originality of the research, declaration is
823 not required.
824 Answer: [Yes]
825 Justification: An earlier pass used a large language model to assign the disclosure-
826 coding categories; those labels were cleared and none enters. The categories now
827 standing were hand-assigned by a human coder from archived release-time docu-
828 ments under the frozen protocol, with an independent second coder blind-coding
829 a provider-stratified 20% sample (Section 5), so no number reported here depends
830 on a model’s judgment. Every number is produced by the released code from the
831 pinned snapshot. Language models were otherwise used only for writing and code
832 assistance.